

Escola Tècnica Superior d'Enginyeria Informàtica

Final Technical Report

Model-Agnostic Interpretability in Predictive Systems

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Abstract

This report applies model-agnostic XAI methods to interpret Random Forest models for bike rental demand and housing prices. By utilizing 1D and 2D Partial Dependence Plots (PDP), we isolate marginal effects such as non-linear climatological impacts and the primary influence of living area on valuation. This approach effectively transforms "black-box" models into transparent tools for informed decision-making.

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1. Introduction

Modern predictive analytics often rely on "black-box" models like Random Forests to achieve high accuracy. However, sometimes, especially for business, accuracy is not enough to make strategic decisions if there is a lack of explanation. Not only the predictive capacity is valuable, but its interpretability: the ability to explain the impact of variables like climate conditions or geographic location.

With this in mind, this report evaluates housing price dynamics and bike rental demand performing an analysis based on Partial Dependence Plots to move beyond simple predictions and understand the logic behind the results. This approach turns the "black-box" problems into a clear tool for making more informed decisions.

2. Bike Rental Demand Analysis

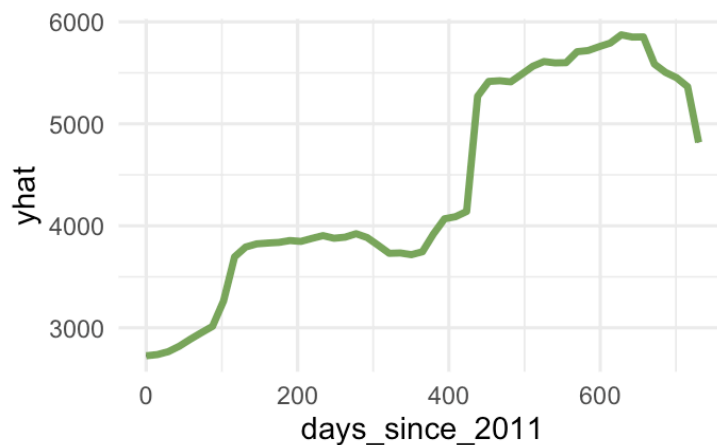


Figure 1: PDP of days_since_2011

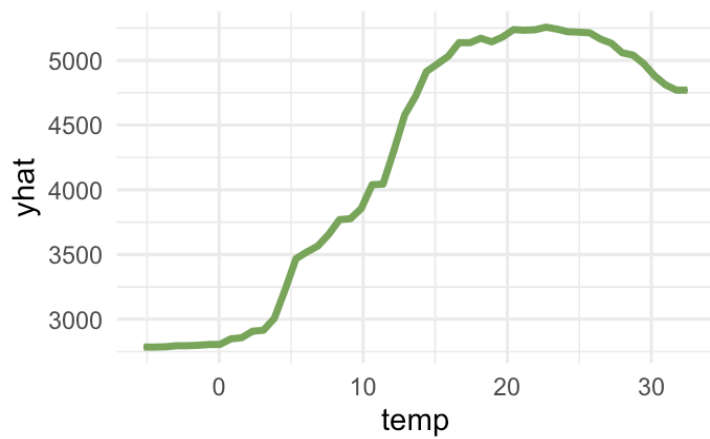


Figure 2: PDP of temperature

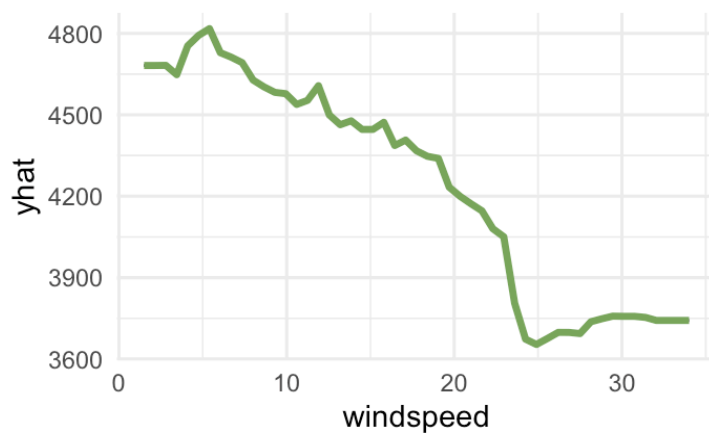


Figure 3: PDP of windspeed

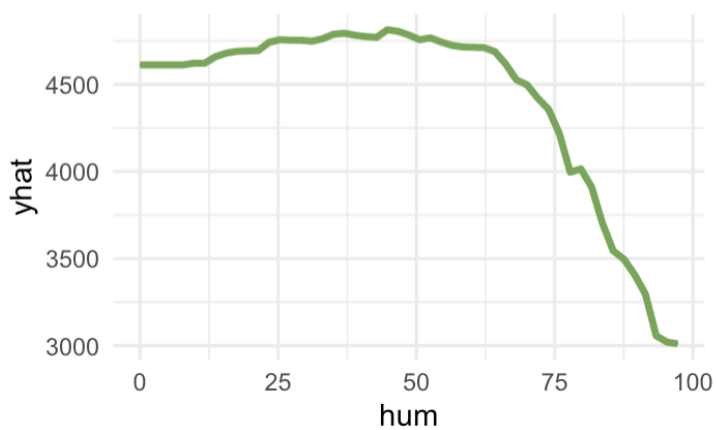


Figure 4: PDP of humidity

Partial Dependence Analysis allows the isolation of the effects of critical variables on rental volume predictions, revealing how the Random Forest model captures the business logic:

- **Days since 2011 (Figure 1):** The plot shows an upward growth trend in demand. Two significant slopes are observed, one around 100 days and another near 420 days, reflecting the service's consolidation.
- **Temperature (Figure 2):** There is a clear positive relationship between temperature and bike usage. Demand surges starting at 5°C and reaches its peak around 25°C, after which a slight drop is observed, indicating that extreme heat discourages cycling.
- **Wind Speed (Figure 3):** Wind speed has a near-linear negative influence. As wind increases, demand decreases, with a particularly sharp drop once it exceeds 20 km/h. It appears that wind is a decisive factor for users.
- **Humidity (Figure 4):** Humidity maintains a neutral to slightly positive effect until it exceeds 65%, at which point rental predictions drop drastically. This suggests the model identifies high humidity levels as a proxy for a high probability of precipitation.

4. Climatological Interactions

To analyze the interaction between multiple variables and their combined effect on bike rental demand, a **Bidimensional Partial Dependence Plot** was generated. Due to the computational intensity of calculating predictions across a 2D grid, a set of random samples was extracted from the dataset to generate the PDP data (Figure 5).

The 2D plot reveals that the highest demand occurs when temperatures are warm (20-25°C) and humidity is relatively low (below 50%). It seems like high humidity levels drastically reduce predicted rentals even when the temperatures are at their best.

Regarding the marginal density plots, the top curve of the temperature indicates that temperature data is most concentrated between 15°C and 25°C, which aligns with rental's prediction peak. The curve on the right illustrates the distribution of humidity levels, with a peak suggesting that most observations fall between 50% and 75%. To summarize, marginal plots provide a context by showing where the model's predictions are supported by the

highest density of real-world data, ensuring that interpretation align with the most frequent weather patterns experienced in the dataset.

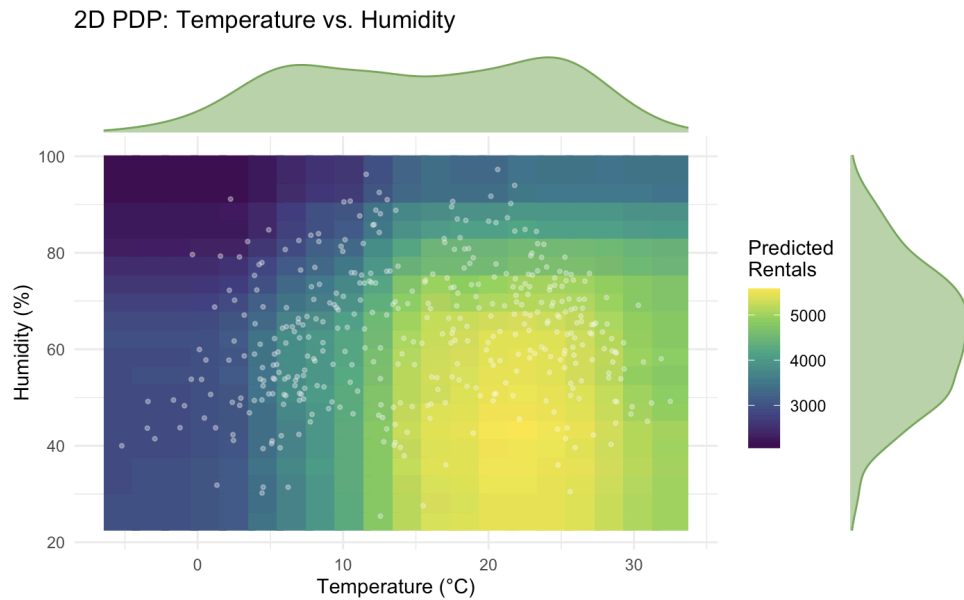


Figure 5: 2D PDP of Humidity vs. Temperature

5. Housing Market Valuation

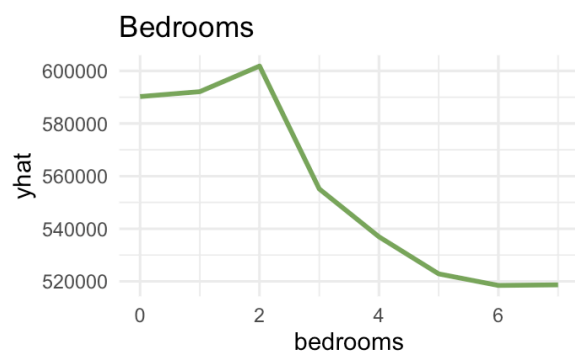


Figure 6: PDP of

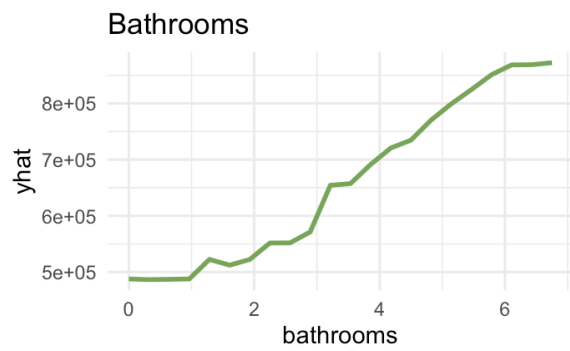


Figure 7: PDP of

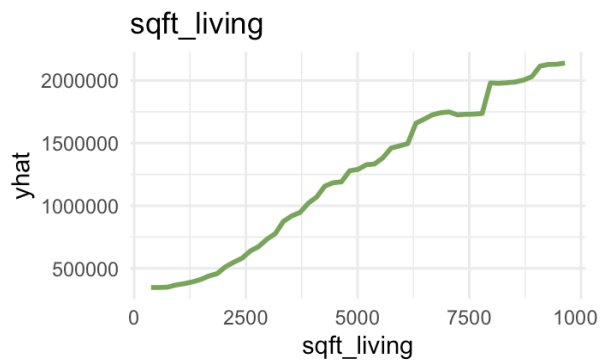


Figure 8: PDP of

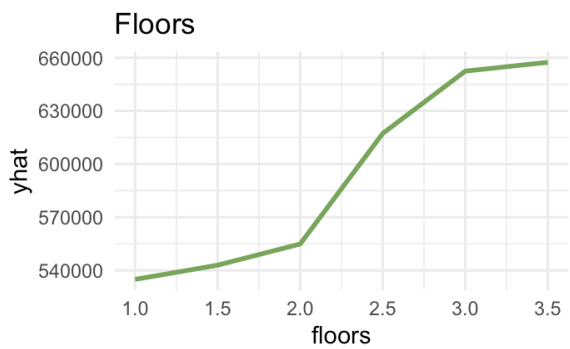


Figure 9: PDP of

In order to analyse the drivers of housing prices, a random forest model was fitted using the `kc_house_data` dataset. The target variable is the sale price, and the main explanatory features considered are the number of bedrooms and bathrooms, the living area in square feet

(sqft_living), the lot size (sqft_lot), the number of floors and the year the house was built. To reduce computational cost and follow the exercise instructions, the model was trained on a random subsample of the available records.

The one-dimensional Partial Dependence Plots reveal that the model learns a clearly increasing relationship between sqft_living and the predicted price: as the living area grows, the expected price rises sharply, especially beyond medium-sized houses. This confirms that usable indoor space is one of the most influential factors for the model. The PDPs for bedrooms and bathrooms also show positive effects, but with more of a stepwise pattern: moving from very few to a reasonable number of rooms or bathrooms increases the predicted price, while adding additional rooms above a certain point has a more limited impact.

In contrast, the effect of floors is more moderate and almost flat in some ranges, suggesting that the number of storeys is less important than size or equipment. Overall, the PDPs indicate that the random forest bases its decisions mainly on the functional capacity and comfort of the property (size and number of bathrooms), while other aspects such as the lot size or the exact number of floors play a secondary role in determining the predicted price.

6. Conclusions